Business Model— Phishing Email Detection System

1. Customer Segments

- Small/medium businesses (SMBs) needing phishing protection but lacking enterprise-grade email filters
- Universities / educational institutions (student/staff emails)
- IT departments & system administrators (corporate)
- Individual power-users (freelancers, home offices) who want advanced email security
- Cybersecurity training / awareness providers

2. Value Propositions

- Hybrid detection: combines rule-based URL & header checks + ML — more robust than naive spam filters
- Real-time analysis & verdict (Safe / Suspicious / Phishing) with **explainable reason & confidence score** — useful for audit & security awareness
- Persistent logging & audit trail — useful for compliance, forensics, and review
- Web dashboard + quarantine interface — user-friendly monitoring, not just a backend filter
- Easy deployment — lightweight, cross-platform, no expensive infrastructure required
- Adaptable/Extensible — new detection rules or ML retraining can be added without redoing entire system

3. Channels

- GitHub / open-source repository for developers and testers
- Web demo / web interface for end-users (hosted SaaS or self-hosted)
- Direct licensing or deployment to small businesses / universities
- Word-of-mouth among cybersecurity communities / developer forums
- Integration or plugins for existing mail servers / organizations

4. Customer Relationships

- Self-service: users install and run the system themselves (self-hosted), minimal support needed
- Documentation(setup, training, usage) — for developers and admins
- Optional offering: managed service / support / consulting for enterprises (on-boarding, custom rules, model retraining)
- Feedback & update loop: users report false positives/negatives → allow retraining / tuning (improves detection over time)

5. Revenue Streams

- **Subscription / Licensing fee** — monthly or yearly for businesses/universities using the system as SaaS(Software as a Service) or managed service
- **One-time deployment / customization fee** — for on-premise installation, integration, custom rule writing
- **Consulting / Support services** — help with configuration, model retraining, logs/quarantine management
- **Training & awareness packages** — using your system + user training for employees / students

6. Key Resources

- The detection engine: ML model + rule-based chain (codebase)
- Skilled developers / maintainers (Python, ML, security experts)
- Infrastructure for hosting (if offering SaaS): server, storage (for logs & quarantine), networking
- Data (email datasets for training / testing / improving model)
- Documentation, support materials

7. Key Activities

- Model maintenance: retraining, tuning, updating with new phishing techniques

- Rule updates: updating URL/header rules to catch new suspicious patterns (domains, TLDs, obfuscation, etc.)
- Web interface & API maintenance (frontend + backend)
- Logging, quarantine and audit trail management (ensuring reliability and security)
- Customer support and consulting (if offering managed service)
- Security updates & monitoring (detecting bypasses, new threats)

8. Key Partnerships

- Email service providers / hosting companies — to integrate your detection engine as a module/plugin
- Universities / educational institutions — pilot users, testers, feedback providers
- Cybersecurity communities / forums — for threat intelligence, collaborative improvement, feedback loop
- Cloud / hosting platforms — for SaaS deployment (if you offer hosted version)
- Compliance/legal advisors — for data protection and privacy compliance (if storing logs or quarantining emails)

9. Cost Structure

- Development & maintenance costs (developer time, bug fixes, feature updates)
- Hosting / infrastructure (servers, storage for logs/quarantine, bandwidth)
- Customer support / consulting overhead (for managed service clients)
- Data acquisition / storage / model training costs (if collecting large datasets)
- Ongoing security audits / updates
- Marketing / documentation / platform maintenance (if offering public SaaS)

Summary

The **Phishing Email Detection System** addresses the growing need for robust email security among SMBs, universities, and individual users — especially those lacking

enterprise-grade email filtering. By combining a hybrid detection engine with a user-friendly web dashboard, persistent logging, and quarantine capability, the system offers both high detection accuracy **and** transparency through confidence scores and rationale.

Revenue can be generated via subscriptions, licensing, consulting, and support — while keeping costs manageable thanks to open-source foundations, modular architecture, and optional self-hosting. Key activities include continuous model/rule maintenance, interface upkeep, and user support. Strategic partnerships with email providers, hosting companies, and cybersecurity communities can expand reach and strengthen threat intelligence.